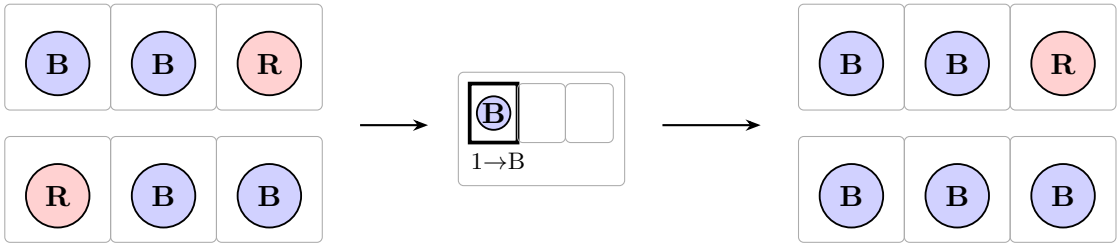
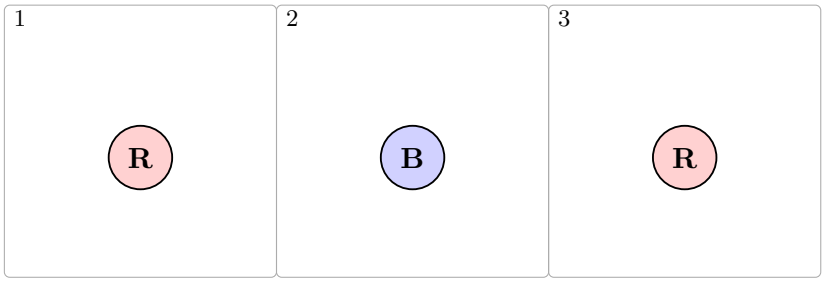
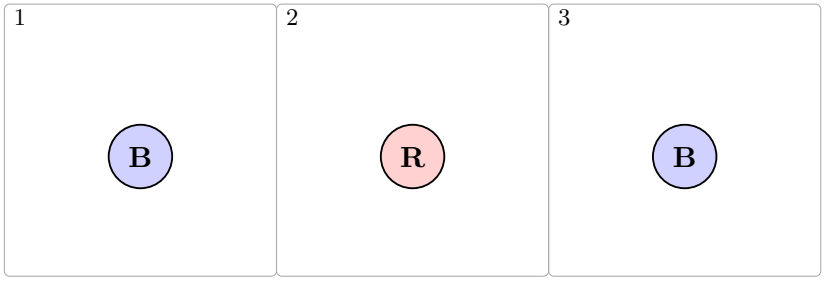
Each instruction names a slot and a color. Set that slot to that color on both boards, even if it is already that color. Slots are 1, 2, 3 from left to right.

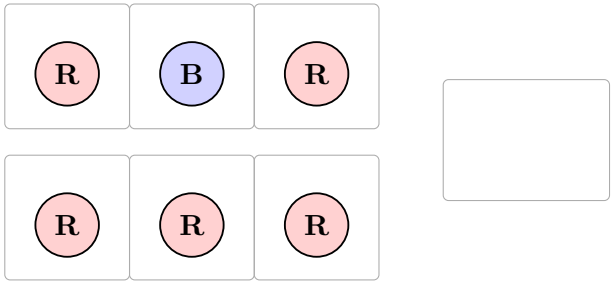


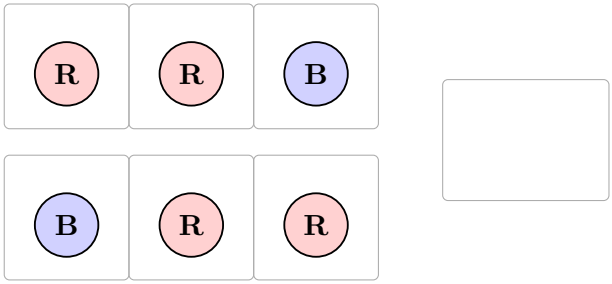

**Problem 1:** Make these boards match using as few instructions as you can. Can any other starts need more instructions?

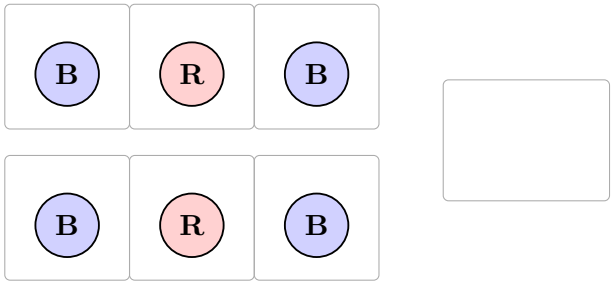


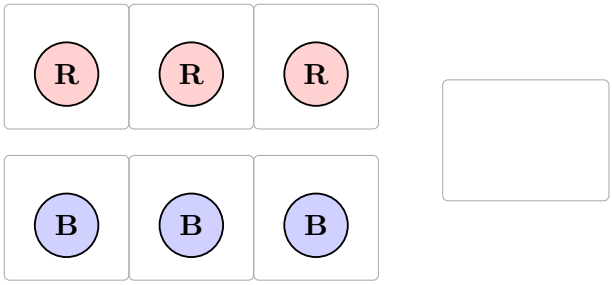



**Problem 2:** How few instructions can make each pair match? Try every pair.

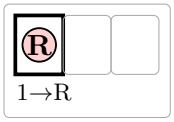


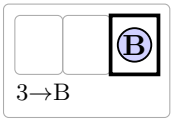



**Problem 4:** Use these two instructions. Make two different starts that end matching, and a pair that ends different.




1→R

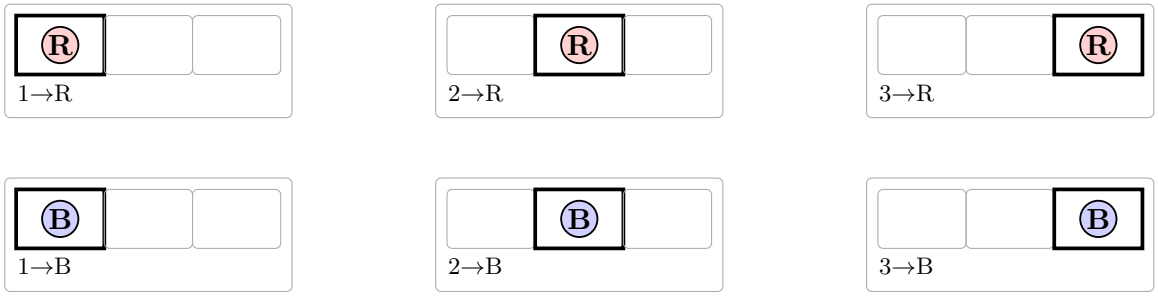



3→B

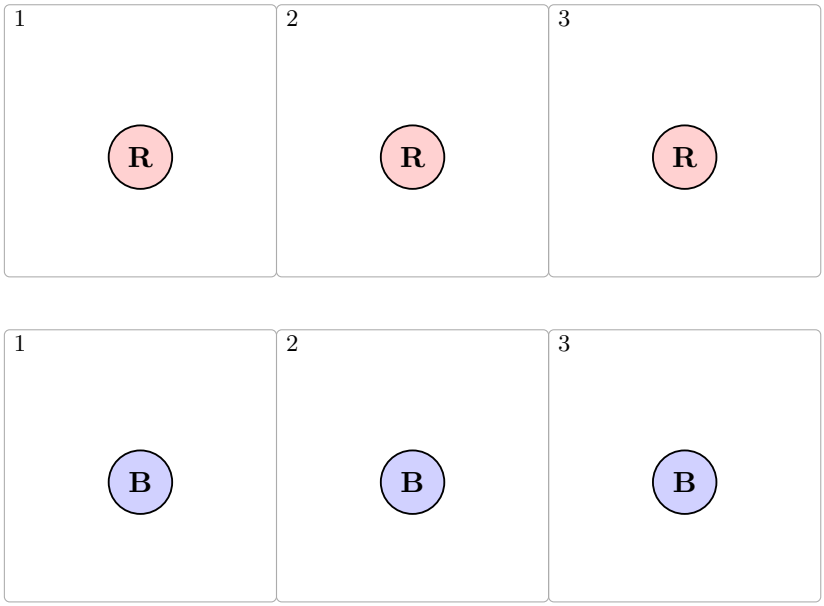




Mix these six cards in a cup. Draw one, use it on both boards, then put it back and mix.




**Problem 6:** Try six draws from these starts, then start again. Could matching take more than three draws?




**Problem 7:** Keep drawing after the boards match. Can any instruction make them different again?